Simulation-based Time Series Analysis:

Testing for Structure, Likelihood-free Estimation, and Prediction with Transformers using Synthetic Data

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Overview of presentation

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Part I

Testing

Economics application

Hardness result

Physical dependence

Type I error control

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Part II

Estimation

Random features

Embedding results

Consistency

Asymptotic normality

Overview of presentation

Part I	Part II	Part III
Testing	**Estimation**	**Prediction**
Economics application	Random features	Foundation models
Hardness result	Embedding results	Sim-and-regression
Physical dependence	Consistency	Conf. distributions
Type I error control	Asymptotic normality	Calibration

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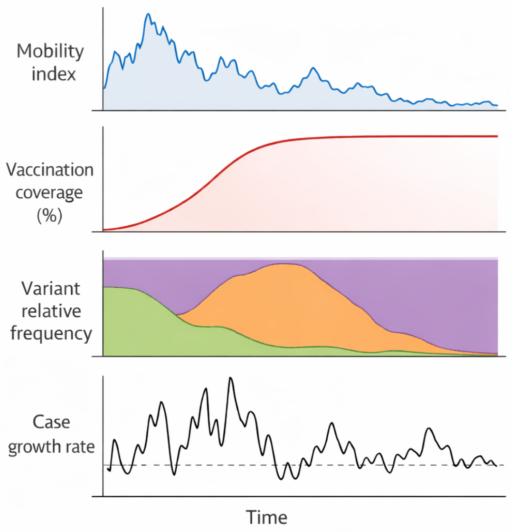
- Example: time series in epidemiology

COVID-19 time series May 2020–2022 (n=365x2=730, d=4)



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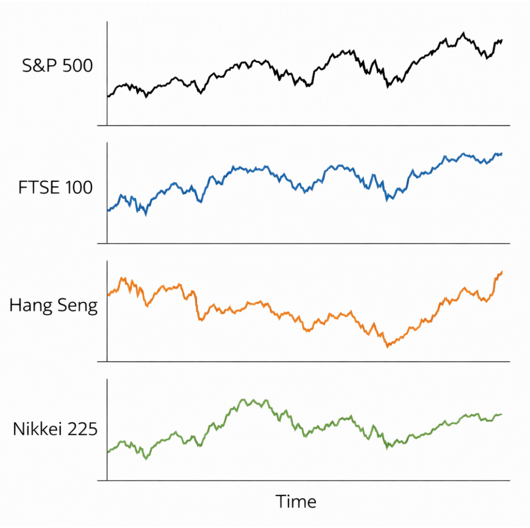
- We show a subset of the results based on BH-adjusted p-values

Closing prices of each stock index over time



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- We reject *every* null hypothesis at the significance level $\alpha = 0.05$

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- Next, we present the testing procedure

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- Calculate test statistic on time series of residual products

$$T(\hat{R}_{1:n}) = \max_{j \in [n]} \left\| \frac{1}{\sqrt{n}} \sum_{t \leq j} \hat{R}_t \right\|_p$$

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- Reject null hypothesis at level α if $T(\hat{R}_{1:n}) > \hat{q}_{1-\alpha}^{\mathrm{boot}}$, else retain

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- **Implication:** We can only hope to achieve Type I error control on a subset of null distributions $\mathcal{P}_n^{\mathrm{CI}}$, so must restrict the null hypothesis to make CI testing feasible
- **Key assumption:** uniform decay of physical dependence measure (Wu 2005) for all time series $X_{1:n}$, $Y_{1:n}$, and $Z_{1:n}$

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- Let $G_t^{(n)} : \mathbb{R}^\infty \times \Theta \to \mathbb{R}^d$ be some measurable function, $t, n, d \in \mathbb{N}$

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- Denote the collections by $\mathcal{P}_n = \{P_{\theta,n} : \theta \in \Theta\}$ and $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$

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Assumption

There exist $\Psi, \rho > 0$ *such that, for some* $q > 2$ *and all* $j \in \mathbb{N}$, *we have*

$$\sup_{\theta\in\Theta}\sup_{\ell\in\mathbb{Z}}\sup_{i\in\mathbb{N}}\sup_{t\in\mathbb{N}} \left\| G_t^{(i)}(\boldsymbol{\epsilon}_\ell, \theta) - G_t^{(i)}(\tilde{\boldsymbol{\epsilon}}_{\ell,j}, \theta) \right\|_{L^q(\theta)} \le \Psi(j \vee 1)^{-\rho}$$
$$\sup_{\theta\in\Theta}\sup_{\ell\in\mathbb{Z}}\sup_{i\in\mathbb{N}}\sup_{t\in\mathbb{N}} \left\| G_t^{(i)}(\boldsymbol{\epsilon}_\ell, \theta) \right\|_{L^q(\theta)} \le \Psi$$

where $\|\cdot\|_{L^q(\theta)} = \mathbb{E}_\theta\left(\|\cdot\|_2^q\right)^{1/q}$, $\mathbb{E}_\theta(\cdot)$ *is expectation w.r.t.* θ*-dependent distribution*

We prove our test has uniformly asymptotic Type I error control

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- Need sufficiently fast convergence rates for *time-varying* regression estimators

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Theorem (Informal)

Recall the test statistic $T(\hat{R}_{1:n})$ and estimated quantile $\hat{q}_{1-\alpha}^{\mathrm{boot}}$. We have

$$\limsup_{n \to \infty} \sup_{P \in \mathcal{P}_n^*} \mathbb{P}_P\left(T(\hat{R}_{1:n}) > \hat{q}_{1-\alpha}^{\mathrm{boot}}\right) \leq \alpha.$$

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- **Key assumption:** We know $G_t^{(n)}$ and can simulate from it at any value of $\theta \in \Theta$

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- For nonstationary time series, need rolling-window estimator (not discussed here)

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- Takes time to find good features
- Can we just use *random features* of the data instead?
- No need to craft features, computationally efficient, little knowledge needed

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- For "almost every" C^1 smooth function Φ from $\mathbb{R}^p$ to $\mathbb{R}^{2p+1}$:
  1. Φ **is one-to-one** on compact subsets of $\mathbb{R}^p$
  2. Φ **has a nice derivative** on certain compact subsets of $\mathbb{R}^p$

  – Details: the $(2p+1) \times p$ Jacobian matrix of Φ has full rank p on each compact subset of a smooth manifold contained in $\mathbb{R}^p$

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- Unless we are very unlucky in our selection of φ, then Φ will one-to-one on Θ and have a nice derivative
- If physical dependence measure (Wu 2005) decays uniformly over Θ, we can estimate $\Phi(\theta)$, $\theta \in \Theta$, via time-averages of $2p+1$ random features

$$\frac{1}{n-m}\sum_{t=m+1}^{n} \varphi\left([X_t(\theta), \ldots, X_{t-m}(\theta)]^\top\right)$$

Example: Mean of Gaussian

- **Goal:** Estimate $\theta_0 = 0$ given $X_1^{\mathrm{obs}}, \ldots, X_n^{\mathrm{obs}} \overset{\mathrm{iid}}{\sim} \mathcal{N}(\theta_0, 1)$ with $\Theta = [-3, 3]$

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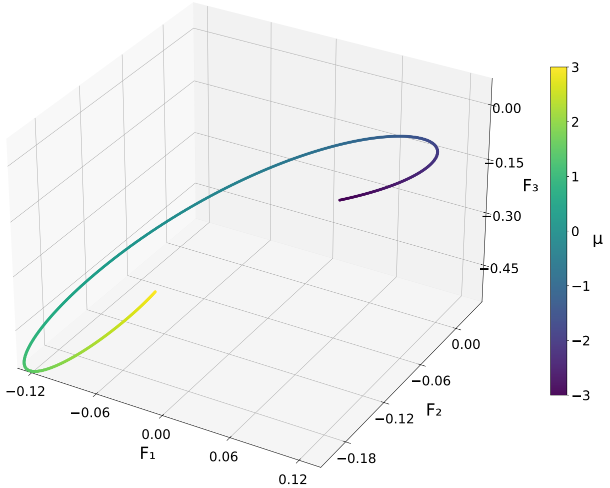
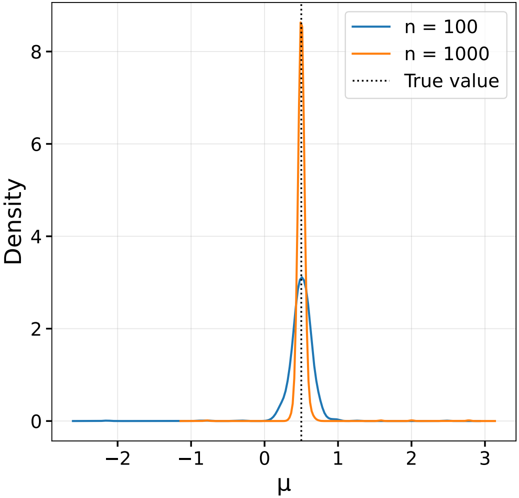
- We will compare the MSE of our estimator $\hat{\theta}$ with $\hat{\theta}^{\mathrm{MLE}} = \frac{1}{n} \sum_{t=1}^{n} X_t$

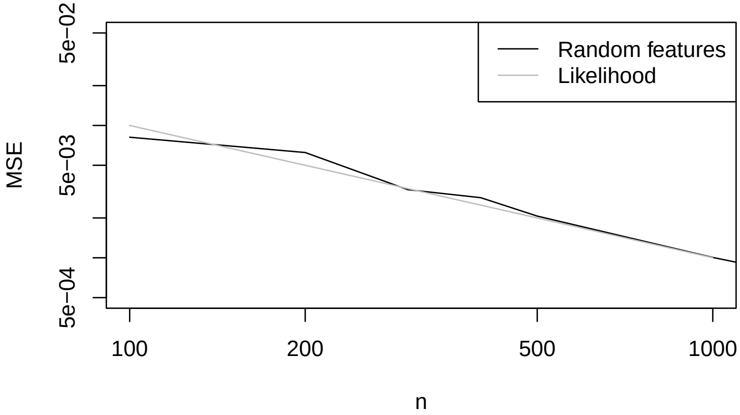
Time-average of 3 features, colored by parameter value ($n = 1000$, $s = 10$)



Density plot based on $1,000$ independent estimates



Theoretical MSE of MLE vs MSE of our estimator (100 replicates per n)



Theoretical guarantees for two estimators

- We prove consistency $\hat{\theta} \xrightarrow{p} \theta_0$ and asymptotic normality

$$\sqrt{n}\left(\hat{\theta} - \theta_0\right) \Rightarrow N\left(0, (\Gamma^\top \Gamma)^{-1} \Gamma^\top V \Gamma (\Gamma^\top \Gamma)^{-1}\right)$$

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- We also prove the same guarantees for a rolling-window estimator
- Examples of nonstationary time series:
  1. ODEs with *temporally dependent* observational noise
  2. State-space models with controls (epidemiology)
  3. Trend-seasonality-noise decomposition with latent process (forecasting)

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- Statistical foundations are unclear, especially for nonstationary time series
- We develop theory, and propose *simulation-and-regression* methods
- In the context of state-space modeling:
  - **Smoothing:** estimate past latent states X with current info
  - **Forecasting:** predict future observations Y or latent states X with current info

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$$\hat{X} = \hat{f}(Y^{\mathrm{obs}})$$

Accounting for parameter uncertainty

- The previous method aims to minimize the Monte Carlo approximation to

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- Why not account for parameter uncertainty?

Our method: Simulation-and-regression with parameter uncertainty

- Let $\widehat{\mathrm{CD}}_n$ be some estimated *confidence distribution* over $\Theta \subset \mathbb{R}^p$ for θ_0

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- Theory of confidence distributions from D. Liu, R. Y. Liu, and Min-ge Xie (2022)
- We propose minimizing the Monte Carlo approximation to

$$R_{P_{\widehat{\mathrm{mix}}}}(f) = \mathbb{E}_{\theta \sim \widehat{\mathrm{CD}}_n}\left[\mathbb{E}_{P_\theta}\left[\sum_{t \in \mathcal{T}} L\left(X_t - [f(Y)]_t\right)\right]\right] = \mathbb{E}_{P_{\widehat{\mathrm{mix}}}}\left[\sum_{t \in \mathcal{T}} L\left(X_t - [f(Y)]_t\right)\right]$$

for some loss L, where the expectation is w.r.t. the mixture distribution $P_{\widehat{\mathrm{mix}}} = \int_\Theta P_\theta d\widehat{\mathrm{CD}}_n(\theta)$ of sequences X, Y corresponding to $\widehat{\mathrm{CD}}_n$

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- When $\widehat{\mathrm{CD}}_n$ is Dirac delta at $\hat{\theta}$, this is like classical simulation-and-regression
- Lastly, we propose the following direction to pursue

Future direction

- Does accounting for parameter uncertainty via our procedure improve the calibration of transformers that output conditional quantiles?

Thank you!

Questions and comments are welcome
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How to get synthetic training data for forecasting with TSFMs

- For forecasting with TSFMs, Dooley et al. (2023) suggest training on simulations from *multiplicative seasonality-trend-noise decomposition models* (many $\theta \in \Theta$)
- We consider an extension with multiplicative autoregressive noise

$$Y_t = \mathrm{Se}\left(t, \theta^{Se}\right) \mathrm{Tr}\left(t, \theta^{Tr}\right) \exp\left(\kappa X_t - \frac{\kappa^2 \sigma_X^2}{2(1-\phi^2)}\right) \exp\left(\varepsilon_t^Y - \frac{\sigma_Y^2}{2}\right)$$
$$X_t = \phi X_{t-1} + \varepsilon_t^X$$

  for some scaling factor $\kappa > 0$ and autoregressive coefficient $\phi \in (-1, 1)$, with $X_0 \sim N(0, \frac{\sigma_X^2}{1-\phi^2})$ and $\varepsilon_t^X \overset{\text{iid}}{\sim} N(0, \sigma_X^2)$, $\varepsilon_t^Y \overset{\text{iid}}{\sim} N(0, \sigma_Y^2)$ for some $\sigma_X, \sigma_Y > 0$
- **Seasonality:** linear combination of sines, cosines (weekly, monthly, yearly periods)
- **Trend:** product of linear trend and exponential trend
- Our rolling-window estimator can be used (consistent and asymptotically normal)

Model we use for smoothing application in epidemiology

- Discrete-valued process driven by a latent autoregressive process and controls
- We condition on the controls, so that we may view each control as a discretely observed function of continuous time $Z_t \equiv Z(t/n)$ that is nice (e.g., Hölder)
- We define a (conditional) state-space model for $(X_{1:n}, Y_{1:n})$ given $Z_{1:n}$ as follows

$$Y_t \mid X_t, Z_t \sim \mathrm{Pois}\left(\exp\left(\kappa X_t + \beta^\top Z_t\right)\right)$$
$$X_t = \mu + \phi(X_{t-1} - \mu) + \varepsilon_t^X$$

- Effect sizes $\kappa \in \mathbb{R}$, $\beta \in \mathbb{R}^{d_Z}$, autoregressive coefficient $\phi \in (-1, 1)$, and mean $\mu \in \mathbb{R}$, with $X_0 \sim N(\mu, \frac{\sigma_X^2}{1-\phi^2})$, $\varepsilon_t^X \overset{\text{iid}}{\sim} N(0, \sigma_X^2)$ indep. of $Z_{1:n}$, for some $\sigma_X > 0$
- $Y_{1:n}$ are observed case counts, $X_{1:n}$ is a *latent infectivity* process, and $Z_{1:n}$ are covariates (e.g., vaccination coverage, variant relative frequency, mobility)
- Our rolling-window estimator can be used (consistent and asymptotically normal)

Calibration

- **Question:** Does accounting for parameter uncertainty improve the calibration of pre-trained/untrained transformers that output conditional quantiles?
- We would like the conditional quantiles to have the correct coverage
- That is, the probability that the time series, at a particular time, is less than or equal to the predicted α-quantile, at that same time, is *actually be* α
- We mainly assess this with synthetic experiments
- For improving calibration with $\widehat{\mathrm{CD}}_n$ *beyond* doing simulation-and-regression and fine-tuning based on the parameter estimate $\hat{\theta}$, we can reduce this to the other questions by instantiating the loss function L as the pinball loss
- For untrained transformers, refer to *simulation-based training*
- For pre-trained transformers, refer to *pre-training*

Calibration

- To assess calibration in the context of quantile forecasting, we adapt the metrics from Adler et al. (2025), which are often based on time-averages
- **Main difference:** We average each metric over s simulations from a parametric model for a *nonstationary* time series
- Let n be the time of forecasting, $h \in [H]$ be the forecasting horizon, $Y_{1:n}$ be the observed time series, $\hat{Y}^q_{n+1:n+h}$ be the predicted quantile for some level $q \in Q = \{0.1, 0.2, \ldots, 0.9\}$, and $\Delta = \{(0.1, 0.9), (0.2, 0.8), (0.3, 0.7), (0.4, 0.6)\}$

Calibration

- For $h \in [H]$, define the *Probabilistic Calibration Error (PCE)* as:

$$\mathrm{PCE}(h) = \frac{1}{|Q|} \sum_{q \in Q} \left| q - \mathbb{I}\left[Y_{n+h} \leq \hat{Y}^{q}_{n+h} \right] \right|$$

- Lower values indicate better-calibrated models:
  1. PCE is lower-bounded by 0 (i.e. predicted quantile always above observation)
  2. PCE is upper-bounded by 0.5 (i.e. predicted quantile always below observation)

Calibration

- Well-calibrated models do not necessarily yield useful forecasts
- A model can predict very large quantiles and be calibrated
- Need metric that captures sharpness (i.e. concentration of predictive distribution)
- We scale the width of a quantile regression-based prediction interval by the width of the corresponding unconditional quantiles estimated via simulation
- For $h \in [H]$, define the *Scaled Interval Width (SIW)* as:

$$\mathrm{SIW}(h) = \frac{1}{|\Delta|} \sum_{(q_{\mathrm{low}}, q_{\mathrm{high}}) \in \Delta} \frac{\hat{Y}_{n+h}^{q_{\mathrm{high}}} - \hat{Y}_{n+h}^{q_{\mathrm{low}}}}{Y_{n+h}^{q_{\mathrm{high}}} - Y_{n+h}^{q_{\mathrm{low}}}}$$

- Lower (resp. higher) values correspond to tighter (resp. wider) prediction intervals

Calibration

- We introduce a metric to help check whether a model is under or over confident
- For $h \in [H]$, define the *Centered Calibration Error (CCE)* as:

$$\mathrm{CCE}(h) = \frac{1}{|\Delta|} \sum_{(q_{\mathrm{low}}, q_{\mathrm{high}}) \in \Delta} \left((q_{\mathrm{high}} - q_{\mathrm{low}}) - \mathbb{I}\left[\hat{Y}_{n+h}^{q_{\mathrm{low}}} \leq Y_{n+h} \leq \hat{Y}_{n+h}^{q_{\mathrm{high}}} \right] \right)$$

- Positive CCE values mean that more observations are outside the prediction interval than expected
- When combined with a low SIW value, this suggests a model is overconfident, i.e. that the prediction intervals are too narrow

Simulation-based training

- **Question:** In the context of simulation-and-regression, does accounting for parameter uncertainty achieve better performance than using parameter estimate?
- When does this inequality hold:

$$\mathbb{E}_{P_{\theta_0}}\left[\sum_{t\in\mathcal{T}} L\left(X_t - [f^*_{P_{\widehat{\mathrm{mix}}}}(Y)]_t\right)\right] < \mathbb{E}_{P_{\theta_0}}\left[\sum_{t\in\mathcal{T}} L\left(X_t - [f^*_{P_{\hat{\theta}}}(Y)]_t\right)\right]$$

  where both expectations are with respect to the distribution P_{θ_0} of the sequences corresponding to the unknown true parameter θ_0, $f^*_{P_{\widehat{\mathrm{mix}}}}$ is the minimizer in the function class of $R_{P_{\widehat{\mathrm{mix}}}}(f)$, and $f^*_{P_{\hat{\theta}}}$ is the minimizer in the function class of $R_{P_{\hat{\theta}}}(f)$

Pre-training

- **Question:** When does a pre-trained transformer fine-tuned with our method outperform a transformer trained from scratch with our method?
- Intuitively, this occurs when the following conditions hold
- Large parameter uncertainty, e.g., when the sample size of the observed time series is small relative to the noise
- The distributions used in the pre-training phase are close to the true distribution, e.g., when we have domain knowledge or we have previously observed time series with similar distributions

Connections of proposed method to Bayesian prediction

- When using the Gaussian depth-CD $N(\hat{\theta}, \hat{\Sigma}/n)$, our proposed method can be viewed as a frequentist, likelihood-free analogue of the Laplace approximation to the posterior distribution in Bayesian statistics
- Specifically, when using the Laplace approximation to the posterior distribution to obtain an approximate posterior predictive distribution
- Recall that the Laplace approximation uses a Gaussian distribution centered at the maximum a posteriori (MAP) estimate with covariance given by the inverse observed Fisher information to approximate the posterior distribution
- The Bernstein–von Mises theorem is used to justify the Laplace approximation
- In contrast, to justify our approach, we use asymptotic theory for likelihood-free parameter estimators for parametric time series models, nonstationary time series theory (Mies 2023), and the theory of confidence distributions

Confidence distributions

- In general, our frequentist procedure can be used with any *confidence distribution* over $\Theta \subset \mathbb{R}^p$ which expresses our uncertainty about the $\theta_0 \in \mathbb{R}^p$
- Ideally, this distribution is easy to simulate from
- We give the definition of confidence distributions for the $p = 1$ case, and the generalization to the $p \in \mathbb{N}$ case
- In the definition, $\Theta \subset \mathbb{R}^p$ is the parameter space and $\mathcal{X}_n$ is the sample space corresponding to the data $X_{1:n}$

Confidence distributions

Definition (Confidence Distribution, $p = 1$, (Min-ge Xie and Singh 2013))

A function $\widehat{\mathrm{CD}}_n(\cdot) = \widehat{\mathrm{CD}}_n(X_{1:n}, \cdot)$ on $\mathcal{X}_n \times \Theta \to [0,1]$ is called a confidence distribution (CD) for a parameter θ, if:

1. For each given $X_{1:n}$, $\widehat{\mathrm{CD}}_n(\cdot) = \widehat{\mathrm{CD}}_n(X_{1:n}, \cdot)$ is a CDF on Θ
2. At the true parameter value $\theta = \theta_0$, $\widehat{\mathrm{CD}}_n(\theta_0) \equiv \widehat{\mathrm{CD}}_n(X_{1:n}, \theta_0)$, as a function of the sample $X_{1:n}$, follows the uniform distribution $U[0,1]$

- Also, the function $\widehat{\mathrm{CD}}_n(\cdot)$ is an asymptotic confidence distribution (aCD), if the $U[0,1]$ requirement is true only asymptotically

Confidence distributions

- The definition of confidence distributions for the general $p \in \mathbb{N}$ case involves depth functions
- As discussed in R. Y. Liu, Parelius, and Singh (1999), a data depth is a way of measuring how deep or central a point $x \in \mathbb{R}^d$ is with respect to a probability distribution P on $\mathbb{R}^d$, $d \in \mathbb{N}$, or with respect to a given data cloud $X_{1:n}$
- In general, a point with a larger depth value means that it is closer to the center of the distribution or more central in a data cloud, while points with smaller depth values are farther away from the center, so data depth is a center-outward ordering of multivariate observations

Confidence distributions

- For example, the Mahalanobis depth (D_M) (Mahalanobis 1936) at $x \in \mathbb{R}^d$ with respect to P is defined as

$$D_M(P, x) = \frac{1}{1 + (x - \mu_P)^\top \Sigma_P^{-1} (x - \mu_P)},$$

where μ_P and Σ_P are the mean vector and the dispersion matrix of P, respectively

- We obtain the sample version of D_M by replacing μ_P and Σ_P by the corresponding sample estimates $\hat{\mu}_P$ and $\hat{\Sigma}_P$
- See R. Y. Liu, Parelius, and Singh (1999) for more examples of data depth

Confidence distributions

Definition (Confidence Distribution, $p \in \mathbb{N}$, (D. Liu, R. Y. Liu, and Min-ge Xie 2022))

A function $\widehat{\mathrm{CD}}_n(\cdot) \equiv \widehat{\mathrm{CD}}_n(X_{1:n}, \cdot)$ on $\Theta \subset \mathbb{R}^p$ is called a depth confidence distribution (depth-CD) associated with depth function D for a vector-valued parameter θ, if:

1. It is a distribution function on the parameter space Θ for any fixed sample set $X_{1:n}$
2. The $(1-\alpha)$ "central region" of the distribution $\widehat{\mathrm{CD}}_n(\cdot)$,

$$\mathcal{R}_{1-\alpha}(\widehat{\mathrm{CD}}_n) = \{\vartheta \in \Theta : C_{\widehat{\mathrm{CD}}_n}(\vartheta) \geq \alpha\},$$

   is a confidence region for θ with a coverage probability of $(1-\alpha)$. Here, the centrality function associated with depth D and confidence distribution $\widehat{\mathrm{CD}}_n(\cdot)$, that is, $C_{\widehat{\mathrm{CD}}_n}(\vartheta)$, is also referred to as a depth confidence curve (depth-CV)

- If the statements in (2) hold only asymptotically, then we refer to $\widehat{\mathrm{CD}}_n$ and $C_{\widehat{\mathrm{CD}}_n}$ as asymptotic depth-CD and asymptotic depth-CV, respectively

Confidence distributions

- In this work, we focus on asymptotic Gaussian depth confidence distributions of the form $\widehat{\mathrm{CD}}_n = N(\hat{\theta}, \hat{\Sigma}/n)$ for a multi-dimensional parameter $\theta \in \mathbb{R}^p$, $p \in \mathbb{N}$
- See Example 3 in D. Liu, R. Y. Liu, and Min-ge Xie (2022) for an illustration with a bivariate normal distribution
- We justify this Gaussianity through the asymptotic normality of the parameter estimator, nonstationary time series theory (Mies 2023), and the following result from D. Liu, R. Y. Liu, and Min-ge Xie (2022) (Theorem 2 therein)

Confidence distributions

Theorem (Depth-CDs from Pivot Statistics (D. Liu, R. Y. Liu, and Min-ge Xie 2022))

Assume that $A_n(X_{1:n})$ is a nonsingular matrix such that $A_n(X_{1:n})(\hat{\theta} - \theta)$ follows a distribution Q_n that is free of all unknown parameters. Also assume that η_n is a random vector, independent of the sample $X_{1:n}$, following the distribution Q_n. Then, conditional on $X_{1:n}$, the distribution function of $(\hat{\theta} - A_n(X_{1:n})^{-1}\eta_n)$ is a depth-CD for θ, under the following conditions:

1. *The depth D is affine-invariant.*
2. *The depth contours $\{\eta_n : D_{Q_n}(\eta_n) = \epsilon\}$ all have probability zero with respect to Q_n for any $\epsilon > 0$.*

Confidence distributions

- Alternatively, use parametric bootstrap (Efron 1979; Efron and Tibshirani 1994)
- To use a parametric bootstrap approach, we would estimate the parameter on the observed data, simulate from the model with the estimated parameter, then re-estimate the parameter for each simulated realization
- The sampling distribution of the re-estimates is used as the depth-CD
- Theorem 1 in D. Liu, R. Y. Liu, and Min-ge Xie (2022) shows that this bootstrap distribution is an asymptotic depth-CD under certain regularity conditions
- For more info about the connections between bootstrap distributions and confidence distributions, see Example 3 in Min-ge Xie and Singh (2013)

Confidence distributions

- We opt for an asymptotic Gaussian depth-cd, rather than a bootstrap depth confidence distribution, to improve the computational efficiency of our method
- This computational efficiency is particularly important, because our proposed method is intended to be applicable to models with intractable likelihoods
- In this setting, parameter estimation typically relies on simulation-based procedures that are substantially more computationally demanding than methods like maximum likelihood estimation
- Therefore, our procedure shares some of the motivations behind the Krinsky and Robb (1986) method

What do we mean by "almost every" function? Prevalent subset

Definition (J. Yorke and Ott (2005))

Let V be a completely metrizable topological vector space. A Borel set $E \subset V$ is said to be **prevalent** if there exists some Borel measure μ on V such that:

1. $0 < \mu(C) < \infty$ for some compact subset C of V
2. The set $E + v$ has full μ-measure (that is, the complement of $E + v$ has measure zero) for all $v \in V$

- In this literature, it is common to say that, if $E \subset V$ contains a prevalent Borel set, then "almost every" element of V lies in E
- In this sense, "almost every" C^1 smooth map from $\mathbb{R}^p$ to $\mathbb{R}^{2p+1}$ has the properties from the embedding theorem of Sauer, J. A. Yorke, and Casdagli (1991)

Definition of smooth manifold (University of Toronto Lecture Notes)

Definition

A C^∞-manifold (i.e. smooth manifold or differentiable manifold) is a Hausdorff topological space M, with countable basis, together with a C^∞-atlas

- The next slides explain the definitions needed to understand this

Definition of smooth manifold (University of Toronto Lecture Notes)

- An n-dimensional manifold is a space that locally looks like $\mathbb{R}^n$. To give a precise meaning to this idea, our space first of all has to come equipped with some topology (so that the word "local" makes sense)
- Recall that a *topological space* is a set M, together with a collection of subsets of M, called *open subsets*, satisfying the following three axioms: (i) the empty set $\emptyset$ and the space M itself are both open, (ii) the intersection of any finite collection of open subsets is open, (iii) the union of any collection of open subsets is open
- The collection of open subsets of M is also called the topology of M. A map $f : M_1 \to M_2$ between topological spaces is called *continuous* if the pre-image of any open subset in M_2 is open in M_1. A continuous map with a continuous inverse is called a *homeomorphism*

Definition of smooth manifold (University of Toronto Lecture Notes)

- One ingredient in the definition of a manifold is that our topological space comes equipped with a covering by open sets which are homeomorphic to open subsets of $\mathbb{R}^n$
- Let M be a topological space. An n-dimensional *chart* for M is a pair (U, ϕ) consisting of an open subset $U \subset M$ and a continuous map $\phi : U \to \mathbb{R}^n$ such that ϕ is a homeomorphism onto its image $\phi(U)$. Two such charts (U_α, ϕ_α), (U_β, ϕ_β) are C^∞-compatible if the transition map

$$\phi_\beta \circ \phi_\alpha^{-1} : \phi_\alpha(U_\alpha \cap U_\beta) \to \phi_\beta(U_\alpha \cap U_\beta)$$

  is a diffeomorphism (a smooth map with smooth inverse). A covering $\mathcal{A} = (U_\alpha)_{\alpha \in A}$ of M by pairwise C^∞-compatible charts is called a C^∞-atlas

Definition of smooth manifold (University of Toronto Lecture Notes)

- We recall that a topological space is called *Hausdorff* if any two points have disjoint open neighborhoods.
- A *basis* for a topological space M is a collection $\mathcal{B}$ of open subsets of M such that every open subset of M is a union of open subsets in the collection $\mathcal{B}$.
- A topological space with countable basis is also called *second countable*
- For example, the collection of open balls $B_\varepsilon(x)$ in $\mathbb{R}^n$ define a basis. But one already has a basis if one takes only all balls $B_\varepsilon(x)$ with $x \in \mathbb{Q}^n$ and $\varepsilon \in \mathbb{Q}_{>0}$; this defines a countable basis.

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